

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (E&C) (Credit System) Degree Examinations**

January 2023

20ECE81 – COMPUTER ARCHITECTURE

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1.	a) What are condition code flags? Explain the four commonly used flags.	10	L*2	1	1
	b) Briefly explain the connection between processor and memory. And also explain the steps of execution of a program.	10	L2	1	1
2.	a) Perform multiplication of $-11 * 6$ using Booth's algorithm.	10	L3	2	1
	b) Draw the Flowchart and algorithm of Booth's algorithm and Restoring division algorithm.	10	L1	1	1
3.	a) Using Non-restoring division, find the remainder and quotient by dividing $7/5$.	10	L3	2	1
	b) Represent -307.1875_{10} in single precision and double precision formats.	10	L3	2	1
Unit – II					
4.	a) Differentiate between Flash Drives and Hard-Disk drives.	10	L1	3	1
	b) Briefly explain ROM and its types.	10	L2	3	1
5.	a) A block set - associative cache consists of a total of 256 blocks divided into 4 block sets. The main memory contains 4096 blocks, each consists of 128 words.				
	i) How many bits are there in a main memory address?				
	ii) How many bits are there in each of the TAG, SET and WORD fields?				
	iii) What is the size of cache memory?	10	L3	3	1
	b) What is Cache memory and explain the working of Cache memory.	10	L2	3	1
6.	a) Discuss the different hazards of pipelining.	10	L2	4	1
	b) Explain the concept of virtual memory.	10	L2	4	1
Unit – III					
7.	a) Briefly explain the concept of Exceptions and its types.	10	L2	5	1
	b) Explain Interrupt Nesting and briefly explain how to handle two or more simultaneous requests.	10	L2	5	1
8.	a) Briefly explain the concept of DMA.	10	L2	5	1
	b) Briefly explain SCSI Bus.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
